# ENGINEERING MATHEMATICS I (EM201T / NEM201T)
## VIBRANT MASTER FORMULA SHEET & COMPLETE PYQ GUIDE
**Complete 5-Unit Coverage · Mid-Sem, SEE & Re-Exam PYQs · Calculator Tricks · 10-Page Exam Companion**

## UNIT I: Partial Differentiation, Multivariable Calculus & Extrema

### 1.1 Limits & Continuity in Higher Dimensions

**A. Limits of Functions of Two Variables** $f(x,y)$: $\lim_{(x,y)\to(a,b)} f(x,y) = L$ means $f(x,y)$ approaches $L$ along **every possible path** approaching $(a,b)$ in the $xy$-plane.

#### Core Limit Theorems & Indeterminate Forms

- **Direct Substitution**: If $f(x,y)$ is continuous at $(a,b)$, then $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.
- **Indeterminate Form $\frac{0}{0}$ (Factorization / Cancellation)**:

$$\lim_{x\to a} \frac{x^n - a^n}{x-a} = na^{n-1}, \quad \lim_{x\to a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m}a^{n-m}$$

- **Two-Path Test for Non-Existence**: If $f(x,y) \to L_1$ along path $y = m_1 x$ and $f(x,y) \to L_2$ along $y = m_2 x$ with $L_1 \neq L_2$, the limit **DOES NOT EXIST**.

#### ⋆ EXAM PYQ 1.1A: Factorization Limit [Mid-Sem 2024 Set B Q1 & 2025 Set B Q1]

**Question**: Evaluate $\lim_{x\to 5} \frac{x-5}{x^2-25}$.
**Step-by-Step Solution**: 1. Direct substitution gives $\frac{5-5}{25-25} = \frac{0}{0}$ (Indeterminate). 2. Factor the denominator: $x^2 - 25 = (x-5)(x+5)$. 3. Cancel common factor $(x-5)$: $\lim_{x\to 5} \frac{1}{x+5} = \frac{1}{5+5} = \frac{1}{10} = \mathbf{0.1}$.

#### ⋆ EXAM PYQ 1.1B: Trigonometric Limit [Assignment Q6 & Q10]

**Question**: Evaluate (i) $\lim_{x\to 0} \frac{\sin 5x}{4x}$ (ii) $\lim_{\theta\to 0} \frac{\sin 8\theta}{\tan 3\theta}$.
**Solution**: (i) Multiply and divide by 5: $\frac{5}{4} \cdot \lim_{x\to 0} \frac{\sin 5x}{5x} = \frac{5}{4}(1) = \mathbf{\frac{5}{4}}$.
(ii) Divide numerator and denominator by $\theta$: $\frac{\lim \frac{\sin 8\theta}{8\theta} \cdot 8}{\lim \frac{\tan 3\theta}{3\theta} \cdot 3} = \frac{8(1)}{3(1)} = \mathbf{\frac{8}{3}}$.

#### ⋆ EXAM PYQ 1.1C: Piecewise Continuity Parameter [Mid-Sem 2024/2025 B1 & Re-Exam 2026 A1]

**Question (2024/2025)**: Find $A$ such that $f(x) = \begin{cases} A + x & x < 2 \\ 1 + x^2 & x \geq 2 \end{cases}$ is continuous at $x = 2$.
**Solution**: 1. Left Hand Limit (LHL) as $x \to 2^-$: $\lim_{x\to 2^-}(A + x) = A + 2$.
2. Right Hand Limit (RHL) / $f(2)$: $1 + (2)^2 = 5$.
3. For continuity, LHL = RHL $\implies A + 2 = 5 \implies \mathbf{A = 3}$.

**Question (Re-Exam 2026)**: Find $B$ such that $f(x) = \begin{cases} 2x^2 + 3 & x \leq 2 \\ 3x + B & x > 2 \end{cases}$ is continuous at $x = 2$.
**Solution**: $f(2) = 2(2)^2 + 3 = 11$. RHL $= 3(2) + B = 6 + B \implies 6 + B = 11 \implies \mathbf{B = 5}$.

#### CASIO fx-991ES PLUS Pro-Tip: Instant Limit Check

1. Switch to **RADIAN Mode**: `SHIFT` → `MODE` → `4: Rad`.
2. Type expression using `ALPHA` + `)` for $X$: e.g., `(X - 5)/(X^2 - 25)`.
3. Press `CALC`, enter $X = \mathbf{5.0001}$ (or $\mathbf{0.0001}$ for $x \to 0$). Output: $\mathbf{0.09999} \approx \mathbf{0.1}$.

## 1.2 Partial Differentiation & Clairaut's Theorem

**A. First-Order Partial Derivatives**:

$$f_x = \frac{\partial f}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h} \quad \text{(treat } y \text{ as constant)}$$

$$f_y = \frac{\partial f}{\partial y} = \lim_{k\to 0} \frac{f(x,y+k) - f(x,y)}{k} \quad \text{(treat } x \text{ as constant)}$$

**B. Second-Order Partial Derivatives**:

$$\text{Pure Partials: } f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \quad f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

$$\text{Mixed Partials: } f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right), \quad f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$$

### Clairaut's Theorem (Symmetry of Mixed Partials)

If $f(x,y)$ and its partial derivatives $f_x, f_y, f_{xy}, f_{yx}$ are continuous in an open region, then:

$$\frac{\partial^2 \mathbf{f}}{\partial \mathbf{x}\partial \mathbf{y}} = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{y}\partial \mathbf{x}} \quad \text{or} \quad \mathbf{f_{xy} = f_{yx}}$$

### ⋆ EXAM PYQ 1.2A: Partial Derivatives of Polynomial [Mid-Sem 2024/2025/2026 B2 & SEE 2024/2025 B5]

**Question**: If $f(x,y) = x^2 + y^2 + x^2y^2$, find all first-order partial derivatives.
**Step-by-Step Solution**: 1. Differentiate w.r.t $x$ (treating $y$ as constant):

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial x}(y^2) + y^2\frac{\partial}{\partial x}(x^2) = \mathbf{2x + 2xy^2}$$

2. Differentiate w.r.t $y$ (treating $x$ as constant):

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2) + \frac{\partial}{\partial y}(y^2) + x^2\frac{\partial}{\partial y}(y^2) = \mathbf{2y + 2x^2y}$$

### ⋆ EXAM PYQ 1.2B: Mixed Partial Derivative [Mid-Sem 2025 Set B Q3 & 2026 Set B Q3]

**Question**: What is the second-order partial derivative $\frac{\partial^2 f}{\partial x \partial y}$ called?
**Answer**: It is called a **Mixed Partial Derivative**. By Clairaut's theorem, $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.

### ⋆ EXAM PYQ 1.2C: Multivariable Polynomial Partials [SEE 2024 Set A B5 & 2025 Set A B5]

**Question**: If $f(x,y) = 6x^2 + 13xy + 4y^2$, find all first-order partial derivatives.
**Solution**:

$$\frac{\partial f}{\partial x} = 12x + 13y, \quad \frac{\partial f}{\partial y} = 13x + 8y$$

## 1.3 Composite Functions, Chain Rule & Total Differentials

### Multivariable Chain Rule Formulas

1. **Case 1: Single Independent Variable**: If $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$, then:

$$\frac{\mathbf{dz}}{\mathbf{dt}} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}}\frac{\mathbf{dx}}{\mathbf{dt}} + \frac{\partial \mathbf{z}}{\partial \mathbf{y}}\frac{\mathbf{dy}}{\mathbf{dt}}$$

2. **Case 2: Two Independent Variables**: If $z = f(x, y)$ where $x = g(u, v)$ and $y = h(u, v)$, then:

$$\frac{\partial \mathbf{z}}{\partial \mathbf{u}} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \mathbf{u}} + \frac{\partial \mathbf{z}}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \mathbf{u}}, \quad \frac{\partial \mathbf{z}}{\partial \mathbf{v}} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \mathbf{v}} + \frac{\partial \mathbf{z}}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \mathbf{v}}$$

3. **Implicit Differentiation**: If $F(x, y) = C$, then $\frac{\mathbf{dy}}{\mathbf{dx}} = -\frac{\mathbf{F_x}}{\mathbf{F_y}} = -\frac{\partial \mathbf{F}/\partial \mathbf{x}}{\partial \mathbf{F}/\partial \mathbf{y}}$.

### ⋆ EXAM PYQ 1.3A: Linear Composite Function [Mid-Sem 2024/2025 B4 & SEE 2024 A5]

**Question**: If $f(x) = 3x + 4$ and $g(x) = x - 2$, find $f(g(x))$.
**Solution**: Substitute $g(x)$ into $f(x)$: $f(g(x)) = 3(g(x)) + 4 = 3(x - 2) + 4 = 3x - 6 + 4 = \mathbf{3x - 2}$.

### ⋆ EXAM PYQ 1.3B: Total Differential under Constraint [Mid-Sem 2025 C2 & 2026 C2]

**Question**: Find total differential coefficient of $u = x^2y$ w.r.t $x$ given $x^2 + xy + y^2 = 1$.
**Solution**: 1. Total differential $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx} = 2xy + x^2\frac{dy}{dx}$.
2. Differentiate constraint $F(x, y) = x^2 + xy + y^2 - 1 = 0$: $2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{2x+y}{x+2y}$.
3. Substitute $\frac{dy}{dx}$: $\frac{du}{dx} = 2xy + x^2\left(-\frac{2x+y}{x+2y}\right) = \frac{\mathbf{x^2y+2xy^2-2x^3}}{\mathbf{x+2y}}$.

## 1.4 Homogeneous Functions & Euler's Theorem

**Definition**: A function $f(x, y)$ is **homogeneous of degree** $n$ if $f(tx, ty) = t^n f(x, y)$ for any real scalar $t > 0$.

### Euler's Theorem Identities

- **First-Order Euler Identity**: If $u = f(x, y)$ is homogeneous of degree $n$:

$$\mathbf{x}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} + \mathbf{y}\frac{\partial \mathbf{u}}{\partial \mathbf{y}} = \mathbf{nu}$$

- **Second-Order Euler Identity**:

$$\mathbf{x^2}\frac{\partial^2 \mathbf{u}}{\partial \mathbf{x^2}} + \mathbf{2xy}\frac{\partial^2 \mathbf{u}}{\partial \mathbf{x}\partial \mathbf{y}} + \mathbf{y^2}\frac{\partial^2 \mathbf{u}}{\partial \mathbf{y^2}} = \mathbf{n(n-1)u}$$

### ⋆ EXAM PYQ 1.4A: Homogeneous Degree Condition [Mid-Sem 2025 Set B Q2 & 2026 Set B Q2]

**Question**: A function $f(x, y)$ is homogeneous of degree $n$ if:
(a) $f(tx, ty) = f(x, y)$ (b) $f(tx, ty) = t^n f(x, y)$ (c) $f(x, y) = x + y$ (d) $f(x, y) = x^n + y^n$
**Answer**: **(b)** $f(tx, ty) = t^n f(x, y)$.

### ⋆ EXAM PYQ 1.4B: Verification of Euler's Theorem [Assignment Q27]

**Question**: If $u = x^3 + 3x^2y + 3xy^2 + y^3$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3u$.
**Proof**: 1. Test homogeneity: $u(tx, ty) = (tx)^3 + 3(tx)^2(ty) + 3(tx)(ty)^2 + (ty)^3 = t^3(x^3 + 3x^2y + 3xy^2 + y^3) = t^3u$.
2. Degree $n = 3$. By Euler's Theorem, $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu = \mathbf{3u}$. (Proved)

## 1.5 Directional Derivatives & Vector Gradient

### Gradient & Directional Derivative Formulas

1. **Gradient Vector** $\nabla\Phi$: For a scalar field $\Phi(x, y, z)$:

$$\nabla\mathbf{\Phi} = \text{grad } \mathbf{\Phi} = \frac{\partial \mathbf{\Phi}}{\partial \mathbf{x}}\mathbf{i} + \frac{\partial \mathbf{\Phi}}{\partial \mathbf{y}}\mathbf{j} + \frac{\partial \mathbf{\Phi}}{\partial \mathbf{z}}\mathbf{k}$$

2. **Unit Vector** $\hat{\mathbf{u}}$: For a direction vector $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$: $\hat{\mathbf{u}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\mathbf{v}}{\sqrt{\mathbf{v_1^2}+\mathbf{v_2^2}+\mathbf{v_3^2}}}$.
3. **Directional Derivative**: $\mathbf{D_{\hat{u}}\Phi} = \nabla\mathbf{\Phi} \cdot \hat{\mathbf{u}} = |\nabla\mathbf{\Phi}| \cos\theta$.
4. **Maximum Rate of Change**: Occurs in the direction of $\nabla\Phi$, with maximum value $= |\nabla\mathbf{\Phi}|$.

### ⋆ EXAM PYQ 1.5A: 3D Gradient Vector [SEE 2024 B2, 2025 B2 & Re-Exam 2026 B5]

**Question**: If $\Phi(x, y, z) = 3x^2y - y^3z^2$, find $\nabla\Phi$ at the point $(1, -2, -1)$.
**Solution**: 1. Compute partial derivatives: $\frac{\partial\Phi}{\partial x} = 6xy$, $\frac{\partial\Phi}{\partial y} = 3x^2 - 3y^2z^2$, $\frac{\partial\Phi}{\partial z} = -2y^3z$. 2. Evaluate at $(1, -2, -1)$: $\frac{\partial\Phi}{\partial x} = 6(1)(-2) = \mathbf{-12}$.
$\frac{\partial\Phi}{\partial y} = 3(1)^2 - 3(-2)^2(-1)^2 = 3 - 12 = \mathbf{-9}$.
$\frac{\partial\Phi}{\partial z} = -2(-2)^3(-1) = -2(-8)(-1) = \mathbf{-16}$. 3. $\nabla\Phi(1, -2, -1) = \mathbf{-12i - 9j - 16k}$.

### ⋆ EXAM PYQ 1.5B: Directional Derivative [Mid-Sem 2025 C1 & 2026 C1]

**Question**: Find the directional derivative of $f(x, y, z) = xyz$ at $(1, -1, 1)$ in direction $3\mathbf{i} - 4\mathbf{k}$.
**Solution**: 1. $\nabla f = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$. At $(1, -1, 1)$, $\nabla f = (-1)(1)\mathbf{i} + (1)(1)\mathbf{j} + (1)(-1)\mathbf{k} = -\mathbf{i} + \mathbf{j} - \mathbf{k}$. 2. Unit vector $\hat{\mathbf{u}} = \frac{3\mathbf{i}-4\mathbf{k}}{\sqrt{3^2+0^2+(-4)^2}} = \frac{3\mathbf{i}-4\mathbf{k}}{5}$. 3. Directional Derivative $= \nabla f \cdot \hat{\mathbf{u}} = (-\mathbf{i}+\mathbf{j}-\mathbf{k}) \cdot \left(\frac{3\mathbf{i}-4\mathbf{k}}{5}\right) = \frac{(-1)(3)+(1)(0)+(-1)(-4)}{5} = \frac{-3+4}{5} = \frac{\mathbf{1}}{\mathbf{5}} = \mathbf{0.2}$.

## 1.6 Tangent Planes, Normal Lines & Relative Extrema

### Tangent Plane, Normal Line & Relative Extrema Discriminant

1. **Tangent Plane**: At point $(x_0, y_0, z_0)$ on surface $z = f(x, y)$:

$$\mathbf{z - z_0} = \mathbf{f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}$$

2. **Normal Line Equations**: $\frac{\mathbf{x-x_0}}{\mathbf{f_x(x_0,y_0)}} = \frac{\mathbf{y-y_0}}{\mathbf{f_y(x_0,y_0)}} = \frac{\mathbf{z-z_0}}{\mathbf{-1}}$.
3. **Extrema Classification**: Calculate critical points where $f_x = 0$ and $f_y = 0$. Define:

$$A = f_{xx}, \quad B = f_{xy}, \quad C = f_{yy}, \quad \mathbf{\Delta = AC - B^2}$$

- If $\mathbf{\Delta > 0}$ and $\mathbf{A > 0} \implies$ **Relative Minimum** at critical point.
- If $\mathbf{\Delta > 0}$ and $\mathbf{A < 0} \implies$ **Relative Maximum** at critical point.
- If $\mathbf{\Delta < 0} \implies$ **Saddle Point** (neither max nor min).
- If $\mathbf{\Delta = 0} \implies$ **Test Inconclusive**.

### ⋆ EXAM PYQ 1.6A: Relative Extrema Classification [SEE 2024 B3, 2025 B3 & Re-Exam 2026 B6]

**Question**: Determine relative extrema of $f(x, y) = 6x^2 + 10y^2 + 2x - 3y + 15$.
**Step-by-Step Solution**: 1. Critical points: $f_x = 12x + 2 = 0 \implies x = -\frac{1}{6}$. $f_y = 20y - 3 = 0 \implies y = \frac{3}{20}$. 2. Second partials: $A = f_{xx} = 12$, $B = f_{xy} = 0$, $C = f_{yy} = 20$. 3. Discriminant $\Delta = AC - B^2 = (12)(20) - (0)^2 = \mathbf{240} > \mathbf{0}$. 4. Since $\Delta > 0$ and $A = 12 > 0$, the function has a **Relative Minimum** at $\left(-\frac{1}{6}, \frac{3}{20}\right)$.

# UNIT II: Infinite Series, Convergence Tests & Expansions

## 2.1 Master Convergence Tests Table

An infinite series $\sum_{n=1}^{\infty} a_n$ converges if its sequence of partial sums $S_N = \sum_{n=1}^{N} a_n$ approaches a finite limit $S$ as $N \to \infty$.

### Summary of Standard Convergence Tests

| Test Name | Mathematical Statement | Conclusion / Condition |
|---|---|---|
| **1. $n$-th Term Test** | $\lim_{n\to\infty} a_n = L$ | If $L \neq 0$, series **DIVERGES**. If $L = 0$, test is inconclusive. |
| **2. Geometric Series** | $\sum_{n=0}^{\infty} ar^n$ | **Converges** if $\lvert r\rvert < 1$ with sum $S = \frac{a}{1-r}$. **Diverges** if $\lvert r\rvert \geq 1$. |
| **3. $p$-Series Test** | $\sum_{n=1}^{\infty} \frac{1}{n^p}$ | **Converges** if $p > 1$. **Diverges** if $p \leq 1$ (e.g., $p = 1$ Harmonic series diverges). |
| **4. D'Alembert's Ratio** | $L = \lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert$ | **Converges** if $L < 1$. **Diverges** if $L > 1$. Inconclusive if $L = 1$. |
| **5. Cauchy's Root Test** | $L = \lim_{n\to\infty} \sqrt[n]{\lvert a_n\rvert}$ | **Converges** if $L < 1$. **Diverges** if $L > 1$. Inconclusive if $L = 1$. |
| **6. Leibniz Test** | $\sum(-1)^{n-1} b_n$ (Alternating) | **Converges** if $b_{n+1} \leq b_n$ for all $n$ AND $\lim_{n\to\infty} b_n = 0$. |

### ⋆ EXAM PYQ 2.1: Finite Series Summation [SEE 2024 Set A B6 & 2025 Set A B6]

**Question**: Find $\sum_{i=1}^{4}(2n+3)$.
**Solution**: Expand term-by-term for $n = 1, 2, 3, 4$: $n = 1 : 2(1) + 3 = 5$, $n = 2 : 2(2) + 3 = 7$, $n = 3 : 2(3) + 3 = 9$, $n = 4 : 2(4) + 3 = 11$.
Sum $= 5 + 7 + 9 + 11 =$ **32**.

## 2.2 Power Series, Taylor & Maclaurin Expansions

**A. Power Series**: $\sum_{n=0}^{\infty} c_n(x-a)^n$. Radius of convergence $R = \lim_{n\to\infty} \left\lvert \frac{c_n}{c_{n+1}} \right\rvert$. Interval of convergence: $(a-R, a+R)$.
**B. Taylor & Maclaurin Series Formulas**:

$$\text{Taylor Series at } x = a: \quad \mathbf{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots}$$

$$\text{Maclaurin Series at } x = 0: \quad \mathbf{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots}$$

### Standard Maclaurin Expansions to Memorize

1. $\mathbf{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots} \quad (-\infty < x < \infty)$
2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad (-\infty < x < \infty)$
3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad (-\infty < x < \infty)$
4. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (-1 < x \leq 1)$
5. $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \quad (\lvert x\rvert < 1)$

### ⋆ EXAM PYQ 2.2: Maclaurin Expansion of $e^x$ [Mid-Sem 2026 Set B B4]

**Question**: Determine the Taylor/Maclaurin series at $x = 0$ for $f(x) = e^x$.
**Step-by-Step Solution**: 1. Compute derivatives at $x = 0$: $f(x) = e^x \implies f(0) = 1$; $f'(x) = e^x \implies f'(0) = 1$; $f''(x) = e^x \implies f''(0) = 1$; $f^{(n)}(0) = 1$. 2. Substitute into Maclaurin formula: $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots$ 3. Result: $\mathbf{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}}$.

# UNIT III: Improper Integrals, Special Functions & Revolutions

## 3.1 Improper Integrals (Type 1 & Type 2)

### Classification of Improper Integrals

- **Type 1 (Infinite Limits of Integration)**: At least one limit is $\pm\infty$.

$$\int_a^{\infty} f(x)dx = \lim_{t\to\infty}\int_a^t f(x)dx, \quad \int_{-\infty}^b f(x)dx = \lim_{t\to-\infty}\int_t^b f(x)dx$$

- **Type 2 (Unbounded Integrands)**: $f(x)$ has an infinite discontinuity at $c \in [a,b]$.

$$\text{If } f(x)\to\infty \text{ at } b: \quad \int_a^b f(x)dx = \lim_{t\to b^-}\int_a^t f(x)dx$$

## 3.2 Beta & Gamma Special Functions

### Beta & Gamma Definitions and Core Identities

1. **Gamma Function Definition**: $\boldsymbol{\Gamma(n) = \int_0^{\infty} x^{n-1}e^{-x}dx}$ $(n>0)$.
2. **Gamma Recurrence Relation**: $\boldsymbol{\Gamma(n+1) = n\Gamma(n)}$. For integer $n$: $\boldsymbol{\Gamma(n+1) = n!}$.
3. **Half-Integer Gamma Value**: $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$.
4. **Negative-Argument Evaluation**: $\boldsymbol{\Gamma(n) = \frac{\Gamma(n+1)}{n}}$.
5. **Beta Function Definition**: $\boldsymbol{B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx}$ $(m>0, n>0)$.
6. **Beta Symmetry**: $\boldsymbol{B(m,n) = B(n,m)}$.
7. **Beta-Gamma Fundamental Relation**: $\boldsymbol{B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}}$.
8. **Trigonometric Form of Beta**: $\boldsymbol{B(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta\cos^{2n-1}\theta\, d\theta}$.

### ⋆ EXAM PYQ 3.2A: Proof of Beta Symmetry [Re-Exam 2026 Set A C1]

**Question**: Prove that $B(m,n) = B(n,m)$.
**Proof**: 1. Start with definition: $B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx$. 2. Substitute $u = 1-x \implies x = 1-u \implies dx = -du$. 3. Limits change: when $x=0, u=1$; when $x=1, u=0$. 4. $B(m,n) = \int_1^0 (1-u)^{m-1}u^{n-1}(-du) = \int_0^1 u^{n-1}(1-u)^{m-1}du = \boldsymbol{B(n,m)}$. (Proved)

### ⋆ EXAM PYQ 3.2B: Gamma Negative-Argument Evaluation [Re-Exam 2026 Set A C2]

**Question**: If $\Gamma(1.8) = 0.9314$, what is the value of $\Gamma(-2.2)$?
**Step-by-Step Solution**: 1. Use recurrence relation backwards: $\Gamma(n) = \frac{\Gamma(n+1)}{n}$. 2. Step 1: $\Gamma(-2.2) = \frac{\Gamma(-1.2)}{-2.2}$. 3. Step 2: $\Gamma(-1.2) = \frac{\Gamma(-0.2)}{-1.2}$. 4. Step 3: $\Gamma(-0.2) = \frac{\Gamma(0.8)}{-0.2}$. 5. Step 4: $\Gamma(0.8) = \frac{\Gamma(1.8)}{0.8} = \frac{0.9314}{0.8} = \boldsymbol{1.16425}$. 6. Substitute back:
$\Gamma(-0.2) = \frac{1.16425}{-0.2} = -5.82125$.
$\Gamma(-1.2) = \frac{-5.82125}{-1.2} = 4.85104$.
$\Gamma(-2.2) = \frac{4.85104}{-2.2} = \boldsymbol{-2.2050}$.

### ⋆ EXAM PYQ 3.2C: Half-Integer Gamma Value [Re-Exam 2026 Set A C3]

**Question**: Find $\Gamma(5/2)$.
**Solution**: Apply $\Gamma(n+1) = n\Gamma(n)$: $\Gamma(5/2) = \Gamma\left(\frac{3}{2}+1\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\sqrt{\pi} = \boldsymbol{\frac{3\sqrt{\pi}}{4}}$.

## 3.3 Bounded Area & Volume of Solids of Revolution

### Area and Volume of Revolution Formulas

1. **2D Enclosed Area**: Area $= \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{y}\,\mathbf{dx} = \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{f(x)}\,\mathbf{dx}$.
2. **Volume of Solid Revolved About $y$-Axis (Cylindrical Shell Method)**:

$$\mathbf{V = 2\pi \int_{a}^{b} x \cdot f(x)\,dx}$$

3. **Volume of Solid Revolved About $x$-Axis (Disk / Washer Method)**:

$$\mathbf{V = \pi \int_{a}^{b} [f(x)]^2\,dx} \quad \text{or} \quad \mathbf{V = \pi \int_{a}^{b} \left(y_{outer}^2 - y_{inner}^2\right) dx}$$

### ⋆ EXAM PYQ 3.3A: Bounded 2D Area [Re-Exam 2026 B4 & Mid-Sem 2026 B2]

**Question**: Find the area of the region bounded by $y = x^2$, the $x$-axis, $x = -2$, and $x = 3$.
**Solution**:

$$\text{Area} = \int_{-2}^{3} x^2 dx = \left[\frac{x^3}{3}\right]_{-2}^{3} = \frac{(3)^3 - (-2)^3}{3} = \frac{27 - (-8)}{3} = \mathbf{\frac{35}{3}} \text{ sq. units}$$

### ⋆ EXAM PYQ 3.3B: Volume Revolved About $y$-Axis [SEE 2024 C5, 2025 C5 & Re-Exam 2026 C5]

**Question**: Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and the $x$-axis over $[1, 3]$ about the $y$-axis.
**Step-by-Step Solution**: 1. Use Cylindrical Shell formula about $y$-axis: $V = 2\pi \int_1^3 x \cdot y\,dx = 2\pi \int_1^3 x(x^2)dx = 2\pi \int_1^3 x^3 dx$. 2. Integrate: $V = 2\pi \left[\frac{x^4}{4}\right]_1^3 = 2\pi \left(\frac{3^4 - 1^4}{4}\right) = 2\pi \left(\frac{81-1}{4}\right) = 2\pi(20) = \mathbf{40\pi}$ cubic units.

### ⋆ EXAM PYQ 3.3C: Washer Volume Revolved About $x$-Axis [Mid-Sem 2026 Set B C3]

**Question**: Find the volume of solid generated by revolving region bounded by $y = x^2 + 2$ and $y = x + 4$ about $x$-axis.
**Solution**: 1. Find intersection points: $x^2 + 2 = x + 4 \implies x^2 - x - 2 = 0 \implies (x-2)(x+1) = 0 \implies x = -1, 2$. 2. Washer formula: $V = \pi \int_{-1}^{2} \left[(x+4)^2 - (x^2+2)^2\right] dx = \pi \int_{-1}^{2} \left[(x^2 + 8x + 16) - (x^4 + 4x^2 + 4)\right] dx = \pi \int_{-1}^{2} \left(-x^4 - 3x^2 + 8x + 12\right) dx$. 3. Integrate: $V = \pi \left[-\frac{x^5}{5} - x^3 + 4x^2 + 12x\right]_{-1}^{2} = \mathbf{\frac{117\pi}{5}}$ cubic units.

# UNIT IV: Multiple Integration & Vector Calculus

## 4.1 Double & Triple Integrals (Cartesian & Polar)

### Multiple Integration & Coordinate Change Rules

1. **Cartesian Double Integral**: $\iint_R f(x,y)\,dy\,dx = \int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right) dx$.
2. **Cartesian to Polar Transformation**:

$$\mathbf{x = r\cos\theta}, \quad \mathbf{y = r\sin\theta}, \quad \mathbf{x^2 + y^2 = r^2}, \quad \mathbf{dx\,dy = r\,dr\,d\theta}$$

3. **Triple Integral (3D Volume Integration)**:

$$\iiint_V f(x,y,z)\,dz\,dy\,dx = \int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)\,dz\,dy\,dx$$

### ⋆ EXAM PYQ 4.1A: 2D Double Integral [SEE 2024 Set A B4 & 2025 Set A B4]

**Question**: Evaluate $\int_0^1\int_0^2 y^2x\,dy\,dx$.

**Solution**: 1. Inner integral w.r.t $y$: $\int_0^2 y^2x\,dy = x\left[\frac{y^3}{3}\right]_0^2 = \frac{8}{3}x$. 2. Outer integral w.r.t $x$: $\int_0^1 \frac{8}{3}x\,dx = \frac{8}{3}\left[\frac{x^2}{2}\right]_0^1 = \frac{8}{3}\left(\frac{1}{2}\right) = \mathbf{\frac{4}{3}}$.

### ⋆ EXAM PYQ 4.1B: 3D Triple Integral [SEE 2024 Set A C1 & 2025 Set A C1]

**Question**: Evaluate $\int_0^2\int_1^3\int_1^2 xy^2z\,dz\,dy\,dx$.

**Step-by-Step Solution**: 1. Integrate w.r.t $z$: $\int_1^2 z\,dz = \left[\frac{z^2}{2}\right]_1^2 = \frac{4-1}{2} = \frac{3}{2}$. 2. Integrate w.r.t $y$: $\int_1^3 y^2\,dy = \left[\frac{y^3}{3}\right]_1^3 = \frac{27-1}{3} = \frac{26}{3}$. 3. Integrate w.r.t $x$: $\int_0^2 x\,dx = \left[\frac{x^2}{2}\right]_0^2 = \frac{4}{2} = 2$. 4. Multiply results: Total Integral $= \left(\frac{3}{2}\right)\times\left(\frac{26}{3}\right)\times(2) = \mathbf{26}$.

### ⋆ EXAM PYQ 4.1C: Polar Double Integral Area [SEE 2024 C4, 2025 C4 & Re-Exam 2026 C4]

**Question**: Find the area bounded between the circles $r = 2\sin\theta$ and $r = 4\sin\theta$.

**Step-by-Step Solution**: 1. Set up Polar Double Integral for Area: Area $= \int_0^\pi \int_{2\sin\theta}^{4\sin\theta} r\,dr\,d\theta$. 2. Inner integral w.r.t $r$: $\int_{2\sin\theta}^{4\sin\theta} r\,dr = \left[\frac{r^2}{2}\right]_{2\sin\theta}^{4\sin\theta} = \frac{16\sin^2\theta - 4\sin^2\theta}{2} = 6\sin^2\theta$. 3. Outer integral w.r.t $\theta$: Area $= \int_0^\pi 6\sin^2\theta\,d\theta = 6\int_0^\pi \frac{1-\cos 2\theta}{2}\,d\theta = 3\left[\theta - \frac{\sin 2\theta}{2}\right]_0^\pi = 3(\pi - 0) = \mathbf{3\pi}$.

## 4.2 Geometric Applications: Enclosed Areas by Double Integration

### Standard Enclosed Area Formulas

1. **Area of Ellipse** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: Area $= \pi\mathbf{ab}$.
2. **Area Between Parabola** $y = 4x - x^2$ **and Line** $y = x$: Area $= \mathbf{\frac{9}{2} = 4.5}$.

### ⋆ EXAM PYQ 4.2A: Area of Ellipse [SEE 2024 Set A C2 & 2025 Set A C2]

**Question**: Calculate the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using double integration.

**Proof**: 1. By symmetry, Total Area $= 4\times$ (Quadrant 1 Area). 2. Quadrant 1 limits: $x$ goes from 0 to $a$; $y$ goes from 0 to $b\sqrt{1-\frac{x^2}{a^2}}$. 3. Area $= 4\int_0^a \left(\int_0^{b\sqrt{1-x^2/a^2}} dy\right) dx = 4b\int_0^a \sqrt{1-\frac{x^2}{a^2}}\,dx$. 4. Let $x = a\sin\theta \implies dx = a\cos\theta\,d\theta$: Area $= 4b\int_0^{\pi/2}(a\cos\theta)(a\cos\theta)d\theta = 4ab\int_0^{\pi/2}\cos^2\theta d\theta = 4ab\left(\frac{\pi}{4}\right) = \pi\mathbf{ab}$.

### ⋆ EXAM PYQ 4.2B: Area Between Intersecting Curves [SEE 2024 Set A C3 & 2025 Set A C3]

**Question**: Find area between parabola $y = 4x - x^2$ and line $y = x$.

**Solution**: 1. Intersection points: $4x - x^2 = x \implies x^2 - 3x = 0 \implies x = 0$ and $x = 3$. 2. Area $= \int_0^3\int_x^{4x-x^2} dy\,dx = \int_0^3 (3x - x^2)dx = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \frac{27}{2} - 9 = \mathbf{\frac{9}{2} = 4.5}$.

## 4.3 Vector Calculus: Divergence, Curl & Vector Theorems

### Vector Calculus Definitions & Core Theorems

For a vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$:

1. **Divergence**: $\nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{\partial \mathbf{P}}{\partial \mathbf{x}} + \frac{\partial \mathbf{Q}}{\partial \mathbf{y}} + \frac{\partial \mathbf{R}}{\partial \mathbf{z}}$.
   - If $\nabla \cdot \mathbf{F} = \mathbf{0} \implies \mathbf{F}$ is **Solenoidal**.
2. **Curl**: $\nabla \times \mathbf{F} = \text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$.
   - If $\nabla \times \mathbf{F} = \mathbf{0} \implies \mathbf{F}$ is **Irrotational** (Conservative).
3. **Green's Theorem in Plane**: For closed curve $C$ bounding region $R$:

$$\oint_{\mathbf{C}} (\mathbf{P}\,\mathbf{dx} + \mathbf{Q}\,\mathbf{dy}) = \iint_{\mathbf{R}} \left( \frac{\partial \mathbf{Q}}{\partial \mathbf{x}} - \frac{\partial \mathbf{P}}{\partial \mathbf{y}} \right) \mathbf{dx}\,\mathbf{dy}$$

4. **Stokes' Theorem**: Relates line integral around closed curve $C$ to surface integral over $S$:

$$\oint_{\mathbf{C}} \mathbf{F} \cdot \mathbf{dr} = \iint_{\mathbf{S}} (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\,\mathbf{dS}$$

5. **Gauss's Divergence Theorem**: Relates surface integral over closed surface $S$ to volume integral over $V$:

$$\iint_{\mathbf{S}} \mathbf{F} \cdot \hat{\mathbf{n}}\,\mathbf{dS} = \iiint_{\mathbf{V}} (\nabla \cdot \mathbf{F})\,\mathbf{dV}$$

### ⋆ EXAM PYQ 4.3: Solenoidal Vector Field Test [Re-Exam 2026 B2]

**Question**: If $\mathbf{A} = x^2z\mathbf{i} - 2y^3z^2\mathbf{j} + xy^2z\mathbf{k}$, find $\nabla \cdot \mathbf{A}$ at $(1, -1, 1)$.
**Solution**: 1. $\nabla \cdot \mathbf{A} = \frac{\partial}{\partial x}(x^2z) + \frac{\partial}{\partial y}(-2y^3z^2) + \frac{\partial}{\partial z}(xy^2z) = 2xz - 6y^2z^2 + xy^2$. 2. At $(1, -1, 1)$: $\nabla \cdot \mathbf{A} = 2(1)(1) - 6(-1)^2(1)^2 + (1)(-1)^2 = 2 - 6 + 1 = \mathbf{-3}$.

# UNIT V: Differential Equations (ODEs & PDEs)

## 5.1 First-Order Linear Ordinary Differential Equations

**Standard Form**: $\frac{dy}{dx} + P(x)y = Q(x)$.

### Integrating Factor & General Solution Recipe

1. **Integrating Factor (IF)**: $\text{IF} = \mathbf{e}^{\int \mathbf{P(x)dx}}$.
2. **General Solution**: $\mathbf{y} \cdot (\text{IF}) = \int \mathbf{Q(x)} \cdot (\text{IF})\,\mathbf{dx} + \mathbf{C}$.

### ⋆ SYLLABUS ILLUSTRATION 5.1: First-Order LDE

**Question**: Solve $\frac{dy}{dx} + \frac{2}{x}y = x^3$.
**Solution**: 1. $P(x) = \frac{2}{x}, Q(x) = x^3 \implies \text{IF} = e^{\int \frac{2}{x}dx} = e^{2\ln x} = e^{\ln(x^2)} = \mathbf{x^2}$. 2. General solution: $y \cdot x^2 = \int x^3 \cdot x^2\,dx + C = \int x^5\,dx + C = \frac{x^6}{6} + C \implies \mathbf{y = \frac{x^4}{6} + Cx^{-2}}$.

## 5.2 Second-Order Linear ODEs with Constant Coefficients

**Standard Form**: $a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$. General Solution: $\mathbf{y} = \mathbf{y_{CF}} + \mathbf{y_{PI}}$.

### Complementary Function ($y_{CF}$) Root Classification

Solve Auxiliary Equation $am^2 + bm + c = 0$:

1. **Real & Distinct Roots** ($m_1 \neq m_2$): $\mathbf{y_{CF} = C_1 e^{m_1 x} + C_2 e^{m_2 x}}$.
2. **Real & Equal Roots** ($m_1 = m_2 = m$): $\mathbf{y_{CF} = (C_1 + C_2 x)e^{mx}}$.
3. **Complex Conjugate Roots** ($m = \alpha \pm i\beta$): $\mathbf{y_{CF} = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)}$.

## 5.3 Cauchy-Euler Equations & Legendre Polynomials

### Cauchy-Euler Substitution & Legendre Polynomials

1. **Cauchy-Euler Equation**: $x^2\frac{d^2y}{dx^2} + ax\frac{dy}{dx} + by = f(x)$.
   - Substitution: $\mathbf{x = e^z} \iff \mathbf{z = \ln x}$.
   - Operational replacements: $\mathbf{x\frac{dy}{dx} = Dy}$, $\mathbf{x^2\frac{d^2y}{dx^2} = D(D-1)y}$ $\left(\text{where } D = \frac{d}{dz}\right)$.
2. **Legendre Polynomials** $P_n(x)$: Solutions to $(1-x^2)y'' - 2xy' + n(n+1)y = 0$.
   - **Rodrigues' Formula**: $\mathbf{P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n}$.
   - Values to memorize: $\mathbf{P_0(x) = 1}$, $\mathbf{P_1(x) = x}$, $\mathbf{P_2(x) = \frac{1}{2}(3x^2-1)}$.
   - **Orthogonality**: $\int_{-1}^{1} P_n(x)P_m(x)\,dx = \mathbf{0}$ for $n \neq m$, and $\frac{\mathbf{2}}{\mathbf{2n+1}}$ for $n = m$.

## 5.4 First-Order Partial Differential Equations (PDEs)

### Lagrange's Method for Linear PDEs & Charpit's Method

1. **Lagrange's Linear PDE**: $P(x,y,z)p + Q(x,y,z)q = R(x,y,z)$ $\left(p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}\right)$.
   - **Auxiliary Equations**: $\frac{\mathbf{dx}}{\mathbf{P}} = \frac{\mathbf{dy}}{\mathbf{Q}} = \frac{\mathbf{dz}}{\mathbf{R}}$.
   - Find two independent integrals $u(x,y,z) = c_1$ and $v(x,y,z) = c_2$. Solution: $\mathbf{\Phi(u,v) = 0}$.
2. **Charpit's Method for Non-Linear PDEs** $f(x,y,z,p,q) = 0$:

$$\frac{\mathbf{dp}}{\mathbf{f_x + pf_z}} = \frac{\mathbf{dq}}{\mathbf{f_y + qf_z}} = \frac{\mathbf{dz}}{\mathbf{-pf_p - qf_q}} = \frac{\mathbf{dx}}{\mathbf{-f_p}} = \frac{\mathbf{dy}}{\mathbf{-f_q}}$$

# UNIT VI: CASIO fx-991ES PLUS Calculator & Exam Master Blueprint

## 6.1 Step-by-Step Calculator Shortcuts

### CASIO fx-991ES PLUS Quick-Reference Button Guides

1. **Essential Setup (Radian Mode)**:
   - Press `SHIFT` → `MODE (SETUP)` → `4: Rad`. (Look for small **R** at top).
2. **Evaluating Indeterminate Limits ($\left[\frac{0}{0}\right]$ Form)**:
   - Type expression using fraction key $\left[\frac{\blacksquare}{\square}\right]$ and `ALPHA` + `)` for $X$.
   - Press `CALC`. Enter $X = \mathbf{5.0001}$ for $x \to 5$, or $X = \mathbf{0.0001}$ for $x \to 0$.
   - Press `=`. Toggle decimal/fraction with `S⇔D` button.
3. **Numerical Derivative Verification ($d/dx$ Function)**:
   - Press `SHIFT` + $\int dx$ (activates `d/dx`).
   - Type function inside brackets: e.g., $\cos(X^2)$ using `ALPHA` + `)` for $X$.
   - Set $x = \mathbf{1}$. Press `=`. Output: $\mathbf{-1.6829}$. Matches $-2(1)\sin(1^2) = -1.6829$!
4. **3D Vector Dot Product for Directional Derivatives**:
   - Enter Vector Mode: `MODE` → `8: VECTOR`.
   - Store $\nabla f$ in VctA: Select `1: VctA` → `1: 3D`. Enter components, press `AC`.
   - Store direction in VctB: `SHIFT` + `5` → `2: Data` → `2: VctB` → `1: 3D`. Enter components, press `AC`.
   - Calculate Dot Product: `SHIFT` + `5` → `3: VctA` → `SHIFT` + `5` → `7: Dot` → `SHIFT` + `5` → `4: VctB` → `=`.

## 6.2 60-Mark SEE Exam Master Strategy

### Time Management & High-Yield Marks Blueprint

- **Section C (30 Marks / 50% of Total)**: 3 out of 5 questions (10 Marks each). Spend 90 mins here first! Focus on Ellipse Area ($\pi ab$), Polar Circles ($3\pi$), Triple Integrals (26), and Volume of Revolution ($40\pi$).
- **Section B (20 Marks)**: 4 out of 6 questions (5 Marks each). Spend 50 mins. Focus on Gradient $\nabla\Phi$, Tangent/Normal lines, and Relative Extrema ($\Delta = AC - B^2$).
- **Section A (10 Marks)**: 5 Short Questions (2 Marks each). Spend 25 mins. Solve or verify instantly using CASIO fx-991ES PLUS!